

Neural Network-Based Fault Detection

Architecture, Training, and Performance Evaluation

Seyed Rumaiz

January 18, 2026

Contents

1	Neural Network Model	2
1.1	Model Architecture	2
2	Training Procedure	3
2.1	Model Compilation	3
2.2	Training Strategy	3
3	Results and Evaluation	4
3.1	Classification Performance	4
3.2	Confusion Matrix	4
3.3	Training and Validation Curves	5
4	Overall Test Metrics	6

Chapter 1

Neural Network Model

1.1 Model Architecture

A feedforward Artificial Neural Network (ANN) was designed for multi-class fault detection. The network consists of three fully connected hidden layers using ReLU activation functions. Batch normalization and dropout regularization were applied after each hidden layer to improve training stability and generalization performance. The output layer employs a softmax activation function to produce class probability estimates.

- **Input Layer:** Dimension equal to the number of input features.
- **Hidden Layer 1:** 128 neurons, ReLU activation, Batch Normalization, Dropout (0.2)
- **Hidden Layer 2:** 64 neurons, ReLU activation, Batch Normalization, Dropout (0.2)
- **Hidden Layer 3:** 32 neurons, ReLU activation, Batch Normalization, Dropout (0.2)
- **Output Layer:** Softmax activation with four output neurons corresponding to fault classes.

Chapter 2

Training Procedure

2.1 Model Compilation

The neural network was compiled using the Adam optimizer and categorical cross-entropy loss, which is appropriate for one-hot encoded multi-class targets. Model performance was monitored using accuracy, precision, recall, and multi-label AUC metrics.

2.2 Training Strategy

The network was trained for a maximum of 20 epochs with a batch size of 32. Early stopping was applied based on validation AUC with a patience of 10 epochs. Class weights were used due to severe class imbalance.

Chapter 3

Results and Evaluation

3.1 Classification Performance

Table 3.1 presents the class-wise precision, recall, and F1-score obtained on the test dataset.

Table 3.1: Classification Report for Base Neural Network

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	232,587
1	1.00	1.00	1.00	1,205
2	0.98	0.98	0.98	37,694
3	0.98	1.00	0.99	1,200
Accuracy	0.99			
Macro Average	0.99	0.99	0.99	272,686
Weighted Average	0.99	0.99	0.99	272,686

3.2 Confusion Matrix

Figure 3.1 shows the confusion matrix for the neural network evaluated on the test dataset.

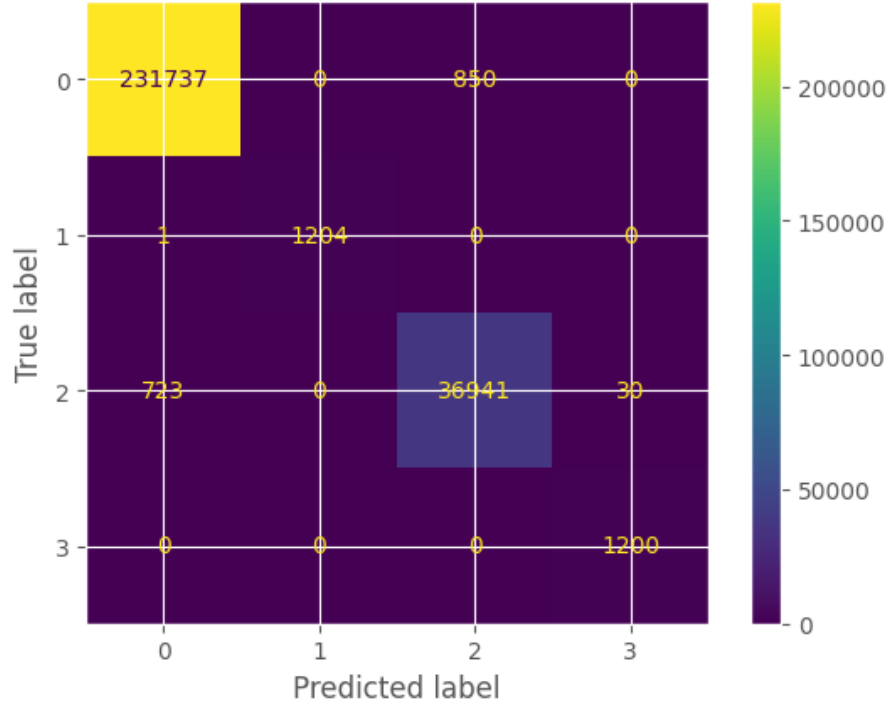


Figure 3.1: Confusion matrix for the neural network classifier.

3.3 Training and Validation Curves

Figure 3.2 illustrates the training and validation behaviour of the neural network across epochs.

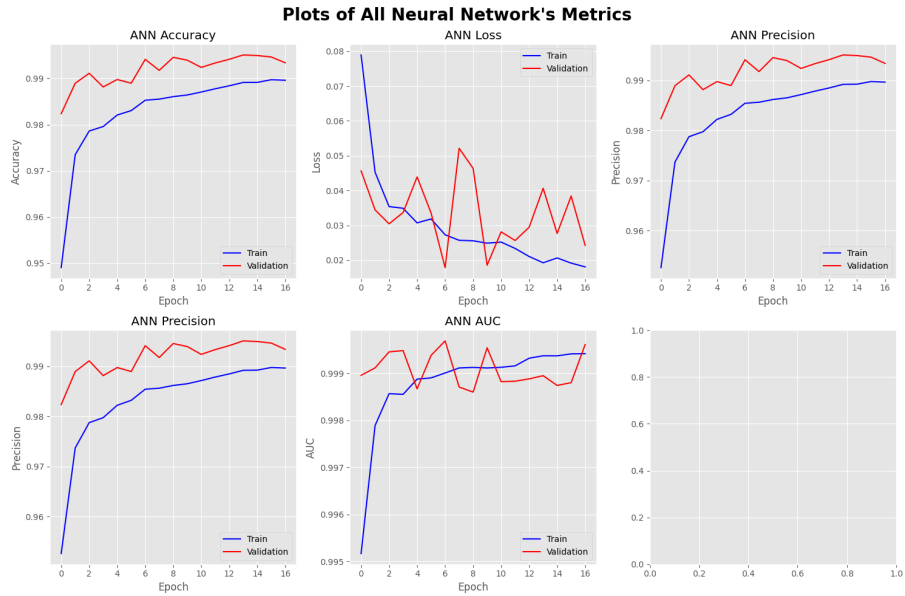


Figure 3.2: Training and validation performance curves across epochs.

Chapter 4

Overall Test Metrics

Table 4.1: Overall Test Set Performance Metrics

Metric	Value
Test Loss	0.018
Test Accuracy	0.994
Test Precision	0.994
Test Recall	0.994
Test AUC	1.000